

Computer Forensics

- Computer Forensics A.A.S. Degree—64 credits

What You Will Learn

MCTC's Information Technology program has three A.A.S. awards: Computer Forensics, Computer Support and Network Administration, and Software Development, as well as a number of diplomas and certificates.

MCTC's Computer Forensics A.A.S. is a unique degree program that combines knowledge and skills of a number of areas including information technology, information security, computer forensics, criminal justice, law, investigation and ethics. Computer forensics professionals are primarily employed by governmental and legal agencies that specialize in cybercrime investigations. Instructors with business and industry experience lead classes and cover the various program areas.

In addition, you will have the chance to work in groups with other class members to learn valuable teamwork and communication skills that reflect the real-world soft-skill requirements. You'll also have the opportunity to participate in various student club activities, technology competitions and industry events.

As a student, you will also have the opportunity to participate in valuable learning activities such as internships, student-learning experiences and the Information Technology Club. Upon completion of the program, you will have gained strong technical problem-solving, interpersonal and communication skills that will be useful as you enter the workforce.

MCTC has articulation agreements with Metropolitan State University, Minnesota State University—Moorehead, St. Mary's University and St. Scholastica for students who wish to pursue a four-year degree.

Where You Could Work

- Help desk centers
- Technical support
- Information technology departments in government and industry
- Network administration
- Internet service providers
- IT consulting firms
- Training centers
- Technology vendors

Computer Forensics A.A.S Degree

Credits: 64

Area 1: ITEC Core (14 credits)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1150	Programming Logic and Design.	3.00
ITEC 1250	Microsoft Windows Operating Systems.	3.00
ITEC 1425	Data Communications.	4.00

Area 2: Criminal Justice/Law Enforcement Core (12 credits)

Course #	Course Name	Credits
PSCI 1101	American Government.	3.00
PSCI 2000	Constitutional Law.	3.00
SOCI 2155	Introduction to Criminal Justice	3.00
LAW 2230	Legal Issues and Law Enforcement	3.00

Area 3: Advanced ITEC/CF Courses (18 credits)

Course #	Course Name	Credits
ITEC 1310	Microcomputer System Maintenance	4.00
ITEC 2710	Microsoft Network Administration	4.00
	OR ITEC 1475 Linux System Administration	
	OR ITEC 2880 Cisco Network Administration	
ITEC 2865	Internet/Intranet Security	4.00
ITEC 2855	Computer Forensics	4.00
ITEC 2950	Information Technology Career Preparation	2.00

Area 4: General Education Courses (20 credits—9 required and 11 electives)

Course #	Course Name	Credits
ENGL 1110	College English 1	3.00
ENGL 1111	College English 2	3.00
SPCH 1000	Fundamentals of Communication.	3.00

General Education Electives (A minimum of 3 credits from MnTC Goal Areas 7, 8, 9 or 10). 11.00

*Enrollment in LAWE courses for non-LAWE majors must be approved by Center for Criminal Justice and Law Enforcement (CCJLE) (an MCTC programs location—St. Paul) director. Please call 651-999-7600.

General Education (20 credits):

To complete the Computer Forensics A.A.S. degree, students are required to successfully complete 20 credits of general education coursework. The required courses are ENGL 1110, ENGL 1111, SPCH 1000 and a minimum of 3 credits from Goal Areas 7, 8, 9 or 10. It is strongly recommended that students seeking transfer into Metropolitan State University's B.A.S. in Computer Forensics complete MATH 1110 and MATH 1150. Program prerequisites: ready for ENGL 1110.

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Courses in this program required achievement of the following academic skill levels:

- **Reading:** To enroll in these program courses, students must have successfully completed READ 0200 or ESOL 0052 or be declared ready to enroll in READ 1300 on the basis of their reading placement test scores.
- **Writing:** To enroll in these program courses, students must have successfully completed ENGL 0900 or ESOL 0051 or be declared ready to enroll in ENGL 1110 on the basis of their writing placement test scores.
- **Math:** To enroll in these program courses, students must have successfully completed MATH 0070 or be declared ready to enroll in MATH 0080 on the basis of their math placement test scores.

Computer Science

- Transfer courses—28 credits

What You Will Learn

Minneapolis Community and Technical College's Computer Science curriculum serves students in three ways:

Elective Credits for the Associate in Arts Degree

If you are seeking an A.A. degree, classes are available to fulfill elective credits and provide additional skills for careers in both computer and business fields.

Transfer Courses

If you are planning to seek a four-year college degree in computer science, engineering, natural sciences or social sciences, the computer programming classes meet the needs of most transfer students who must have introductory programming courses for their majors.

Computer Literacy

Both CSCI 1100: Introduction to Computers and CSCI 1200: Computer Application in Home and Business fulfill the computer literacy requirement for the A.A. degree.

Transfer Courses

Credits: 28

Course #	Course Name	Credits
CSCI 1100	Introduction to Computers	3.00
CSCI 1200	Computer Applications in Home and Business	3.00
CSCI 1500	Introduction to Programming and Problem Solving	3.00
CSCI 1700	Introduction to Computer Programming in Java	3.00
CSCI 1730	Introduction to Object-Oriented Programming With C++.	4.00
CSCI 1901	Structure of Computer Programming 1. .	4.00
CSCI 1902	Structure of Computer Programming 2. .	4.00
CSCI 2011	Discrete Mathematics.	4.00

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to work closely with a counselor or advisor to plan an appropriate program for transfer.

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Computer Software Development

- A.A.S. Degree—64 credits
- Software Developer Diploma—54 credits
- JAVA Developer Certificate—29 credits
- Visual BASIC .NET Developer Certificate—29 credits
- Visual C# Developer Certificate—29 credits
- Web Developer Certificate—29 credits
- Database Specialist Certificate—29 credits
- Open Source Developer Certificate—26 credits (pending MnSCU approval)

Software Development A.A.S. Degree

Credits: 64

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1150	Programming Logic and Design.	3.00
ITEC 1250	Microsoft Windows Operating Systems. .3.00	
ITEC 1425	Data Communications.	4.00

Advanced Courses

(Listed below are additional courses required for completion of this program.)

Course #	Course Name	Credits
ITEC 1375	Database Design and Implementation . .	4.00
*ITEC 2425	Infotech Project Management	4.00
OR		
*ITEC 2775	Systems Analysis and Design	4.00
ITEC 2950	Information Technology Career Preparation	2.00

Choose two sets of languages for a total of 16 credits:

ITEC 1750	Visual BASIC .Net/1	4.00
ITEC 2450	Visual BASIC .Net/2	4.00
ITEC 1771	Java /1	4.00
ITEC 2541	Java /2	4.00
ITEC 1700	Visual C# .NET/1	4.00
ITEC 2400	Visual C# .NET/2	4.00
ITEC 1555	Web Programming	4.00
ITEC 2555	Web Client/Server Programming.	4.00

Advanced credits

General Education Credits (see below)	16.00
Electives Block Credits (see below)	8.00

*Take one of the following: ITEC 2425 **OR** ITEC 2775.

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

Elective Courses (8 credits)

Course #	Course Name	Credits
ITEC 1460	SQL Server Database Design and Implementation	4.00
ITEC 1475	Linux System Administration	4.00
ITEC 1585	E-Commerce Development and Implementation	4.00
ITEC 2075	Application Support and Resource Utilization	4.00
ITEC 2245	Computer Gaming Design and Development.	4.00
ITEC 2855	Computer Forensic.	4.00
ITEC 2900	Information Technology Internship	2.00

General Education Courses (16 credits)

To complete the Computer Support and Network Administration A.A.S. degree, students are required to successfully complete 16 credits of general education coursework. The coursework must be selected from at least three of the MnTC goal areas. A minimum of 3 credits must be completed from Goal Area 1; and 3 credits from Goal Areas 7, 8, 9 or 10. Specific courses that are required are ENGL 1110, ENGL 1111, and SPCH 1000. MATH 1110, MATH 1150 or MATH 1160 is recommended.

Courses in this program require achievement of the following program admissions requirements:

- **Reading:** To enroll in these program courses, students must have successfully completed READ 0200 or ESOL 0052 or be declared ready to enroll in READ 1300 on the basis of their reading placement test scores.
- **Math:** To enroll in these program courses, students must have successfully completed MATH 0070 or be declared ready to enroll in MATH 0080 on the basis of their math assessment test score.

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Software Developer Diploma

Credits: 54

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1150	Programming Logic and Design.	3.00
ITEC 1250	Microsoft Windows Operating Systems. .	3.00
ITEC 1425	Data Communications.	4.00

Advanced Courses

(Listed below are additional courses required for completion of this program.)

Course #	Course Name	Credits
ITEC 1375	Database Design and Implementation . .	4.00
*ITEC 2425	Infotech Project Management	4.00
OR		
*ITEC 2775	Systems Analysis and Design	4.00
ITEC 2950	Information Technology Career Preparation	2.00

*Take one of the following: ITEC 2425 **OR** ITEC 2775.

Choose two sets of languages for a total of 16 credits:

ITEC 1750	Visual BASIC .Net/1	4.00
ITEC 2450	Visual BASIC .Net/2	4.00
ITEC 1771	Java/1	4.00
ITEC 2541	Java/2	4.00
ITEC 1700	Visual C# .NET/1	4.00
ITEC 2400	Visual C# .NET/2	4.00

ITEC 1555	Web Programming	4.00
ITEC 2555	Web Client/Server Programming.	4.00

Advanced credits

General Education Credits (see below)	6.00
Electives Block Credits (see below)	8.00

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

Elective Courses (8 credits)

Course #	Course Name	Credits
ITEC 1460	SQL Server Database Design and Implementation	4.00
ITEC 1475	Linux System Administration	4.00
ITEC 1585	E-Commerce Development and Implementation	4.00
ITEC 2075	Application Support and Resource Utilization	4.00
ITEC 2245	Computer Gaming Design and Development.	4.00
ITEC 2900	Information Technology Internship	2.00

General Education Courses (6 credits)

To complete the Software Developer Diploma, students are required to successfully complete 6 credits of general education coursework. The specific courses that are required are ENGL 1110 and SPCH 1000.

JAVA Developer Certificate

Credits: 29

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1150	Programming Logic and Design.	3.00
ITEC 1250	Microsoft Windows Operating Systems. .	3.00

Advanced Courses

(Listed below are additional courses required for completion of this program.)

Course #	Course Name	Credits
ITEC 1375	Database Design and Implementation . .	4.00
ITEC 1771	Java/1	4.00
ITEC 2541	Java/2	4.00
*ITEC 2425	Infotech Project Management	4.00
OR		
*ITEC 2775	Systems Analysis and Design	4.00
General Education Courses (see below)		3.00

*Take one of the following: ITEC 2425 **OR** ITEC 2775.

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

To complete the Java Developer Certificate, students are required to successfully complete 3 credits of general education coursework from MnTC Goal Area 1.

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Visual BASIC .NET Developer Certificate

Credits: 29

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1150	Programming Logic and Design.	3.00
ITEC 1250	Microsoft Windows Operating Systems. .	3.00

Advanced Courses

(Listed below are additional courses required for completion of this program.)

Course #	Course Name	Credits
ITEC 1375	Database Design and Implementation . .	4.00
ITEC 1750	Visual BASIC .Net/1	4.00
ITEC 2450	Visual BASIC .Net/2	4.00
*ITEC 2425	Infotech Project Management	4.00
OR		
*ITEC 2775	Systems Analysis and Design	4.00
General Education Courses (see below)		3.00

*Take one of the following: ITEC 2425 **OR** ITEC 2775.

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

General Education Courses (3 credits)

To complete the Visual BASIC .Net Developer certificate, students are required to successfully complete 3 credits of general education coursework from MnTC Goal Area 1.

Visual C# Developer Certificate

Credits: 29

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1150	Programming Logic and Design.	3.00
ITEC 1250	Microsoft Windows Operating Systems. .	3.00

Advanced Courses

(Listed below are additional courses required for completion of this program.)

Course #	Course Name	Credits
ITEC 1375	Database Design and Implementation . .	4.00
ITEC 1770	Visual C#.NET/1	4.00
ITEC 2400	Visual C#.Net/2	4.00
*ITEC 2425	Infotech Project Management	4.00
OR		
*ITEC 2775	Systems Analysis and Design	4.00
General Education Courses (see below)		3.00

*Take one of the following: ITEC 2425 **OR** ITEC 2775.

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

General Education Courses (3 credits)

To complete the Visual C# Developer certificate, students are required to successfully complete 3 credits of general education coursework from MnTC Goal Area 1.

Web Developer Certificate

Credits: 29

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1150	Programming Logic and Design.	3.00
ITEC 1250	Microsoft Windows Operating Systems. .	3.00

Advanced Courses

(Listed below are additional courses required for your completion of this program.)

Course #	Course Name	Credits
ITEC 1375	Database Design and Implementation . .	4.00
ITEC 1555	Web Programming	4.00
ITEC 2555	Web Client/ Server Programming	4.00
*ITEC 2425	Infotech Project Management	4.00
OR		
*ITEC 2775	Systems Analysis and Design	4.00
General Education Courses (see below)		3.00

*Take one of the following: ITEC 2425 **OR** ITEC 2775.

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

General Education Courses (3 credits)

To complete the Web Developer certificate, students are required to successfully complete 3 credits of general education coursework from MnTC Goal Area 1.

Database Specialist Certificate

Credits: 29

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1150	Programming Logic and Design.	3.00
ITEC 1250	Microsoft Windows Operating Systems. .	3.00

Advanced Courses

(Listed below are additional courses required for completion of this program.)

Course #	Course Name	Credits
ITEC 1375	Database Design and Implementation . .	4.00
ITEC 1460	SQL Server Database Design and Implementation	4.00
ITEC 1770	Visual C#.Net/1 or Java/1 or VB/1.net (sets of 2)	4.00
ITEC 2400	Visual C#.Net/2 or Java/2 or VB/2.net (sets of 2)	4.00
General Education Courses (see below)		3.00

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

General Education Courses (3 credits)

To complete the Database Specialist certificate, students are required to successfully complete 3 credits of general education coursework from MnTC Goal Area 1.

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Open Source Developer Certificate*

Credits: 26

* Pending MnSCU approval

First Semester

<i>Course #</i>	<i>Course Name</i>	<i>Credits</i>
ITEC 1100	Information Technology Concepts.	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1150	Programming Logic and Design	3.00
ITEC 1475	Linux System Administration	4.00
ENGL 1110	College English 1.	3.00
Total semester credits:		14.00

Second Semester

<i>Course #</i>	<i>Course Name</i>	<i>Credits</i>
ITEC 1771	JAVA Programming Part 1.	4.00
ITEC 2541	JAVA Programming Part 2.	4.00
ITEC 2855	MySQL Database Design and Implementation.	4.00
Total semester credits:		12.00

Please note that the second semester course ITEC 1771 has ITEC 1150 as a prerequisite. Students must complete ITEC 1771 prior to taking ITEC 2541.

Computer Support and Network Administration

- Computer Support and Network Administration A.A.S. Degree—64 credits
- Computer Support and Network Administration Diploma—54 credits
- Information Assurance Professional Diploma—47 credits
- Cisco Network Administrator Certificate—26 credits
- Computer Security Specialist Certificate—26 credits
- Linux Network Administrator Certificate—26 credits
- Microsoft Network Administrator Certificate—26 credits
- Wireless Network Administrator Certificate—26 credits
- Database Administrator Certificate—26 credits

Software Development

- Software Development A.A.S. Degree—64 credits
- Software Developer Diploma—54 credits
- Game Developer Certificate—29 credits
- JAVA Developer Certificate—29 credits
- Visual BASIC .NET Developer Certificate—29 credits
- Visual C# Developer Certificate—25 credits
- Web Developer Certificate—29 credits
- Database Specialist Certificate—29 credits

What You Will Learn

MCTC's Information Technology program has two A.A.S. awards: Computer Support and Network Administration and Software Development, as well as a number of diplomas and certificates.

MCTC's Computer Support and Network Administration degree provides a comprehensive, industry-focused training on network support and Internet technologies. You will learn how to install, configure, and maintain computer hardware, protocols, operating systems, software and Internet-working devices, with special emphasis on developing troubleshooting skills. Instructors with real-world experience lead classes and cover key topics such as network administration, client/server installation and support, Web server management, database implementation, and network security.

In addition, you will have the chance to work in groups with other class members to learn valuable teamwork and communication skills that reflect the real-world soft-skill requirements. You'll also have the opportunity to participate in various student club activities, technology competitions and industry events.

MCTC's Software Development degree will empower you to systematically analyze problems and create software solutions for business, scientific fields and government. You will learn how to create and maintain computer programs using a variety of computer programming languages. You will also study client/server architecture, data communications, data process principles, computer business applications, graphical user interfaces, Web development and e-business/e-commerce concepts.

To prepare programs for implementation, you will learn to design and develop in multiple programming languages. You'll also test and maintain programs to verify proper operation with data and reliability testing, including debugging to correct errors that are encountered. Many of the Information Technology courses are geared toward vendor-based and vendor-neutral industry standard certifications, such as those from Microsoft, Sun and CompTIA.

As a student, you will also have the opportunity to participate in valuable learning activities such as internships, student learning experiences and the Information Technology Club. Upon completion of the program, you will have gained strong problem-solving, interpersonal and communication skills that will be useful as you enter the workforce.

MCTC has an articulation agreement with Metropolitan State University, Minnesota State University—Moorhead, St. Mary's University, and St. Scholastica for students who wish to pursue a four-year degree.

Where You Could Work

- Help desk centers
- Technical support
- Information technology departments in government and industry
- Network administration
- Internet service providers
- IT consulting firms
- Training centers
- Technology vendors
- Software consulting firms
- Application software developers
- Web developers

Students are accepted into this program for fall, spring and summer terms.

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Courses in this program require achievement of the following academic skill levels:

- **Reading:** To enroll in these program courses, students must have successfully completed READ 0200 or ESOL 0052 or be declared ready to enroll in READ 1300 on the basis of their reading placement test scores.
- **Math:** To enroll in these program courses, students must have successfully completed MATH 0700 or be declared ready to enroll in MATH 0080 on the basis of their math assessment test scores.

Computer Support and Network Administration A.A.S. Degree

Credits: 64

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1150	Computer Programming Principles	3.00
ITEC 1250	Microsoft Windows Operating Systems. .3.00	
ITEC 1425	Data Communications.	4.00

Advanced Courses

(Listed below are additional courses required for completion of this program.)

Course #	Course Name	Credits
ITEC 1310	Microcomputer System Maintenance . . .4.00	
*ITEC 1475	Linux System Administration	4.00
OR		
*ITEC 2425	Information Technology Project Management	4.00
ITEC 2710	Microsoft Network Administration	4.00
ITEC 2865	Internet/Intranet Security: Implementation and Management	4.00
ITEC 2880	Cisco Network Administration	4.00
ITEC 2910	LAN Implementation and Management	4.00
ITEC 2950	Information Technology Career Preparation	2.00
General Education Courses (see next column)		16.00
Electives (see next column)		8.00

*Take one of the following: ITEC 2425 **OR** ITEC 1475.

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

General Education Courses (16 credits)

To complete the Computer Support and Network Administration A.A.S. degree, students are required to successfully complete 16 credits of general education coursework. This coursework must be selected from at least three of the MnTC goal areas. A minimum of 3 credits must be completed from Goal Area 1; and 3 credits from Goal Areas 7, 8, 9 or 10. Specific courses that are required are ENGL 1110, ENGL 1111 and SPCH 1000. MATH 1110, MATH 1150 or MATH 1160 is recommended.

Elective Courses (8 credits)

Course #	Course Name	Credits
ITEC 1375	Database Design and Implementation . . .4.00	
ITEC 1460	SQL Server Database Design and Implementation	4.00
ITEC 1555	Web Programming	4.00
ITEC 1585	E-Commerce Development and Implementation	4.00
ITEC 1950	Computer Telephony Integration	4.00
ITEC 2075	Application Support and Resource Utilization	4.00
ITEC 2425	Information Technology Project Management	4.00
ITEC 2775	Systems Analysis and Design	4.00
ITEC 2795	Web Server and TCP/IP Implementation	4.00
ITEC 2820	Wireless LAN/WAN Administration	4.00
ITEC 2855	Computer Forensic	4.00
ITEC 2890	Firewall Implementation and Management	4.00
ITEC 2895	Security Implementation and Risk Analysis	4.00
ITEC 2900	Information Technology Internship	2.00–4.00

Computer Support and Network Administration Diploma

Credits: 54

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1150	Computer Programming Principles	3.00
ITEC 1425	Data Communications.	4.00
ITEC 1250	Microsoft Windows Operating Systems. .3.00	

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Advanced Courses

Listed below are additional courses required for completion of this program. First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

Course #	Course Name	Credits
ITEC 1310	Microcomputer System Maintenance . . .	4.00
ITEC 1475	Linux System Administration	4.00
ITEC 2910	LAN Implementation and Management	4.00
ITEC 2710	Microsoft Network Administration	4.00
ITEC 2865	Internet/Intranet Security: Implementation and Management	4.00
ITEC 2885	Cisco Network Administration	4.00
ITEC 2950	Information Technology Career Preparation	2.00
Electives		8.00
General Education Courses (see below)		6.00

General Education Courses (6 credits)

To complete the Computer Support and Network Administration diploma, students are required to successfully complete 6 credits of general education coursework. A minimum of 3 credits must be completed from Goal Area 1; and 3 credits from Goal Areas 7, 8, 9 or 10.

Elective Courses (8 credits)

Course #	Course Name	Credits
ITEC 1375	Database Design and Implementation . .	4.00
ITEC 1460	SQL Server Database Design and Implementation	4.00
ITEC 1475	Linux System Administration	4.00
ITEC 1555	Web Programming	4.00
ITEC 1585	E-Commerce Development and Implementation	4.00
ITEC 1950	Computer Telephony Integration	4.00
ITEC 2075	Application Support and Resource Utilization	4.00
ITEC 2425	Infotech Project Management	4.00
ITEC 2795	Web Server and TCP/IP Implementation	4.00
ITEC 2820	Wireless LAN/WAN Administration . . .	4.00
ITEC 2855	Computer Forensic	4.00
ITEC 2890	Firewall Implementation and Management	4.00
ITEC 2895	Security Implementation and Risk Analysis	4.00
ITEC 2900	Information Technology Internship	2.00–4.00

Information Assurance Professional Diploma

Credits: 47

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1150	Programming Logic and Design.	3.00
ITEC 1250	Microsoft Windows Operating Systems. .	3.00
ITEC 1425	Data Communications.	4.00

Advanced Courses

(Listed below are additional courses required for completion of this program.)

Course #	Course Name	Credits
ITEC 1310	Microcomputer System Maintenance . .	4.00
*ITEC 2710	MS Network Administration	4.00
OR		
*ITEC 1475	Linux Network Administration	4.00
*ITEC 2875	Cisco Network Administration	4.00
OR		
*ITEC 2795	Web Server and TCP/IP Implementation	4.00
OR		
*ITEC 1375	Database Design and Implementation . .	4.00
OR		
*ITEC 1460	SQL Server Database Design and Implementation	4.00
ITEC 2865	Internet/Intranet Security	4.00
ITEC 2895	Security Implementation and Risk Analysis	4.00
ITEC 2910	LAN Implementation and Management	4.00
General Education Courses (see below)		9.00

*Take one of the following: ITEC 2710 **OR** ITEC 1475.

*Take one of the following: ITEC 2875 **OR** ITEC 2795.

*Take one of the following: ITEC 1375 **OR** ITEC 1460.

Recommended for students interested in transferring into Metropolitan State's B.A.S. in Information Assurance:
MATH 1110—College Algebra
MATH 1150—Statistics

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

General Education Courses (9 credits)

To complete the Computer Support and Network Administration diploma, students are required to successfully complete 9 credits of general education coursework. This coursework must be selected from a minimum of 3 credits from Goal Area 1; and 6 credits from any Goal Area.

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Cisco Network Administrator Certificate

Credits: 26

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1250	Microsoft Windows Operating Systems. .3.00	
ITEC 1425	Data Communications.	4.00

Advanced Courses

(Listed below are additional courses required for your completion of this program)

Course #	Course Name	Credits
*ITEC 2710	MS Network Administration	4.00
OR		
*ITEC 1475	Linux Network Administration	4.00
ITEC 2875	Cisco Network Administration	4.00
ITEC 2880	Firewall Implementation and Management 4.00	
General Education Courses (see below)		3.00

* Take one of the following: ITEC 2710 or ITEC 1475.

First semester courses will generally be selected with the assistance of a counselor or advisor during orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

General Education Courses (3 credits)

To complete the Network Support Professional diploma, students are required to successfully complete 3 credits of general education coursework from MnTC Goal Area 1.

Computer Security Specialist Certificate

Credits: 26

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1250	Microsoft Windows Operating Systems. .3.00	
ITEC 1425	Data Communications.	4.00

Advanced Courses

(Listed below are additional courses required for completion of this program.)

Course #	Course Name	Credits
*ITEC 2710	MS Network Administration	4.00
OR		
*ITEC 1475	Linux Network Administration	4.00
ITEC 2865	Internet/Intranet Security: Implementation and Management	4.00
ITEC 2895	Security Implementation and Risk Analysis	4.00
General Education Courses (see below)		3.00
* Take one of the following: ITEC 2710 OR ITEC 1475.		

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

General Education Courses (3 credits)

To complete the Network Support Professional diploma, students are required to successfully complete 3 credits of general education coursework from MnTC Goal Area 1.

Linux Network Administrator Certificate

Credits: 26

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1250	Microsoft Windows Operating Systems. .3.00	
ITEC 1425	Data Communications.	4.00

Advanced Courses

(Listed below are additional courses required for completion of this program.)

Course #	Course Name	Credits
ITEC 1475	Linux System Administration	4.00
ITEC 2425	Infotech Project Management	4.00
ITEC 2910	LAN Implementation and Management 4.00	
General Education Courses (see below)		3.00

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

To complete the Network Support Professional diploma, students are required to successfully complete 3 credits of general education coursework from MnTC Goal Area 1.

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Microsoft Network Administrator Certificate

Credits: 26

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1250	Microsoft Windows Operating Systems. .3.00	
ITEC 1425	Data Communications.	4.00

Advanced Courses

(Listed below are additional courses required for completion of this program.)

Course #	Course Name	Credits
ITEC 2910	LAN Implementation and Management	4.00
ITEC 2425	Infotech Project Management	4.00
ITEC 2710	Microsoft Network Administration	4.00
General Education Courses (see below)		3.00

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

General Education Courses (3 credits)

Students are required to successfully complete 3 credits of general education coursework from MnTC Goal Area 1.

Wireless Network Administrator Certificate

Credits: 26

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1250	Microsoft Windows Operating Systems. .3.00	
ITEC 1425	Data Communications.	4.00

Advanced Courses

(Listed below are additional courses required for completion of this program.)

Course #	Course Name	Credits
*ITEC 2710	Microsoft Network Administration	4.00
OR		
*ITEC 1475	Linux Network Administration	4.00
ITEC 2425	Infotech Project Management	4.00
ITEC 2820	Wireless LAN/WAN Administration	4.00
General Education Courses (see below)		3.00

* Take one of the following: ITEC 2710 or ITEC 1475.

First semester courses will generally be selected with the assistance of a counselor or advisor during orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

General Education Courses (3 credits)

Students are required to successfully complete 3 credits of general education coursework from MnTC Goal Area 1.

Database Administrator Certificate

Credits: 26

Introductory Courses

(Listed below are required courses we suggest students choose from during their first semester in this program.)

Course #	Course Name	Credits
ITEC 1100	Information Technology Concepts	2.00
ITEC 1110	Information Technology Skills	2.00
ITEC 1250	Microsoft Windows Operating Systems. .3.00	
ITEC 1425	Data Communications.	4.00

Advanced Courses

(Listed below are additional courses required for completion of this program.)

Course #	Course Name	Credits
ITEC 1375	Database Design and Implementation	4.00
ITEC 1460	SQL Server Database Design and Implementation	4.00
ITEC 2425	Infotech Project Management	4.00
General Education Courses (see below)		3.00

First semester courses will generally be selected with the assistance of a counselor or advisor during Orientation. Students are encouraged to visit with a counselor or advisor for planning beyond the first semester in the program.

General Education Courses (3 credits)

Students are required to successfully complete 3 credits of general education coursework from MnTC Goal Area 1.